

Julio Cáceres Gonzales

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SUMMARY

Postdoctoral Scholar at Vanderbilt University who studies quantum symmetries, with a focus on those captured by subfactors. I have ample experience understanding and simplifying complex problems. Through my classroom experience and talks I have also developed strong public speaking skills. Seeking a position as a Data Scientist, Data Analyst of Machine Learning Engineer.

SKILLS & CERTIFICATIONS

- Languages & Platforms: Python, Mathematica, Excel
- Python Libraries: PyTorch, Scikit-learn, Numpy, Sympy, Pandas
- Soft Skills: [Leadership](#), Public Speaking
- Certifications: [The Erdős Institute Data Science Boot Camp](#), [Snowflake Data Warehousing Essentials](#), [Wolfram U Data Science Boot Camp](#)

EXPERIENCE

Vanderbilt University: Nashville, TN **2024 - Present**
Postdoctoral Scholar

- Performed research in subfactor theory. Subfactors capture the quantum symmetries of certain physical systems. My work contributed to the construction of new examples of exotic subfactors with infinite depth, the first of their kind.
- Presented my work as an invited speaker at 2 different conferences, one of them being abroad.
- Developed course material and taught 3 undergraduate courses: Multivariable Calculus, Ordinary Differential Equations, and Linear Algebra with Differential Equations.
- Organized the Directed Reading Program of the Mathematics Department at Vanderbilt, which pairs ungraduated students with graduate students or postdocs to work on a semester long project. Currently working with a student on a project about the Universal Approximation Theorem for Neural Networks.
- Participated in an ongoing research seminar with the goal of developing new invariants for subfactors with infinite depth.

Vanderbilt University: Nashville, TN **2019 - 2024**
Graduate Student

- Performed research in subfactor theory. As part of my thesis, I used Mathematica to perform large searches of graphs with very strict properties. These graphs were then used to construct new exotic subfactors.
- Presented my work as an invited speaker at 7 different conferences at various institutions in the US.
- Organized the Directed Reading Program of the Mathematics Department at Vanderbilt. Mentored three undergraduate students as part of this program.
- Served as Teaching Assistant for 5 calculus courses and as Instructor for 3 calculus courses. Received an award for my teaching.
- Served as a tutor for the Tutoring Center at Vanderbilt. Worked with over 30 students in all levels of calculus.

Banco Central de Reserva del Perú: Lima, Perú **2016**
Internship

- Used Python to improve data preprocessing on country wide credit card information. Reduced the processing time from 2 hours to less than 5 minutes.

SELECTED PROJECTS

Team 10: Predicting Power Outages - The Erdős Institute [\[github\]](#), [\[video\]](#), [\[slides\]](#) **Spring 2025**

- Was awarded top project by The goal of this project was to predict power outages caused by extreme weather events at the county level in different states in the US. We used multiple time series modelling techniques (Linear Regression, XGBoost, etc) and a neural network (LSTM) for our predictions.
 - Skills used: TensorFlow, SKtime, Pandas, GeoPandas, Copernicus API.

EDUCATION

Vanderbilt University, PhD in Mathematics (GPA = 4.00) **2024**
B.F. Bryant Award for Excellence in Teaching, Vanderbilt University **2024**
Harold Stirling Vanderbilt award for outstanding PhD. students, Vanderbilt University **2022**
Russell G. Hamilton Scholar for PhD. Students with outstanding potential, Vanderbilt University **2019**
Universidade Federal de Santa Catarina (Brasil), Master's in Mathematics (GPA = 4.00) **2019**
Universidad Nacional de Ingeniería (Peru), Bachelor of Science (GPA = 4.00) **2016**

PUBLICATIONS

- [Graph planar algebra embeddings and infinite depth subfactors](#), Dietmar Bisch, Julio Cáceres **2024**
- [New hyperfinite subfactors with infinite depth](#), Dietmar Bisch, Julio Cáceres **2025**